

Assignment: A Day in the Life of a Data Analyst

Probability and Statistics with Python

You have just joined as a **Junior Data Analyst** in a growing analytics company. On your very first week, your manager assigns you a set of real-world tasks from different clients. Each task is linked to probability, statistics, and Python. Your goal is to solve them one by one to prove you are ready for the role!

Tasks

Task 1: The Gym Client – “Heights Matter”

A fitness trainer has collected the heights of 15 gym members:

162, 158, 165, 170, 172, 168, 175, 160, 166, 171, 167, 169, 173, 174, 161

The trainer wants to know:

- How the heights are **distributed in groups**.
- A clear **visual histogram** to show his clients.

Use your skills to create a grouped frequency distribution and a histogram.

Task 2: The Retail Client – “Show Me Growth”

A shopkeeper, Meera, recorded sales for 6 months:

Jan=250, Feb=270, Mar=300, Apr=280, May=310, Jun=330

She is preparing for an investor meeting and says: “I don’t want boring tables, I want a story of growth.”

Plot a **line graph** to visualize her sales trend.

Task 3: The School Project – “Do Hours Matter?”

A teacher tracked students’ study hours and scores:

Study Hours	1	2	3	4	5	6	7
Scores	50	55	60	65	70	75	80

She wants to prove that more study hours improve marks.

Create a **scatter plot** to show the relationship.

Task 4: The Mall Project – “Who Buys What?”

A shopping mall hired you to study customer behavior using their dataset (downloaded from Kaggle).

You must:

- Load the dataset and display the first five rows.
- Print the column names and data types.
- Find the number of rows and columns.
- Identify each variable as **Nominal**, **Ordinal**, **Interval**, or **Ratio**.
- Visualize by:
 - A histogram of **Age**
 - A scatter plot between **Annual Income** and **Spending Score**

The manager will use this to design targeted offers.

Task 5: The Game Startup – “Coin Toss Challenge”

A startup is creating a new online game. The first round is a coin toss twice.

Simulate the two tosses and list all possible outcomes.

Task 6: The Dice Café – “Roll It Right”

A café hosting board games wants a **fair dice guide**.

Simulate rolling a die and calculate its **Probability Mass Function (PMF)**.

Task 7: The Factory – “Check My Screws”

In a screw manufacturing factory, the weights of screws follow a **normal distribution**.

Generate a dataset of screw weights and plot both:

- The **PDF (Probability Density Function)**
- The **CDF (Cumulative Distribution Function)**

Task 8: The Production Unit – “Defects or Not?”

A production manager reports the following distribution of defects per batch:

Defects (X)	0	1	2	3
P(X)	0.1	0.3	0.4	0.2

Tasks:

- Check if this is a **valid PMF**.
- Compute the **cumulative probabilities** and plot the **CDF**.

Task 9: The Call Center – “Ticket Trouble”

A call center receives different numbers of support tickets daily.

Explain and model:

- Sample space and events
- Discrete and continuous random variables
- PMF, PDF, and CDF for tickets

Task 10: The Weather Department – “Rain or Shine?”

The meteorological department tracks daily rain/no-rain for 30 days.

Simulate daily weather and present:

- The **PMF** (frequency of rainy vs. dry days)
- The **CDF** (cumulative rainy days over 30 days)

Conclusion

You are the data analyst connecting all these clients through your analysis. Each task builds your reputation. Complete them step by step, and by the end, you will have touched multiple industries using probability, statistics, and Python.